Multi-Robot Ground Texture SLAM with Communication-Efficient Data Sharing

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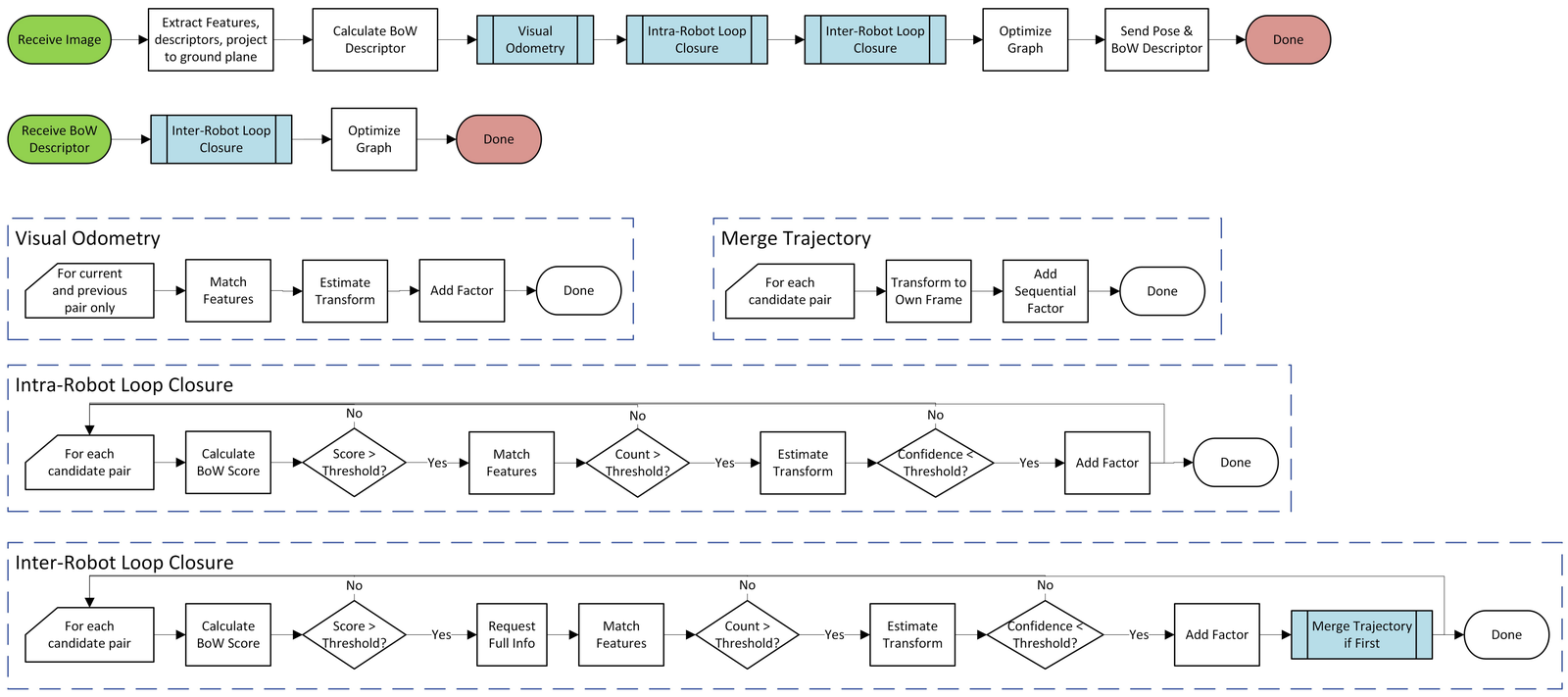
Shipboard Visual SLAM

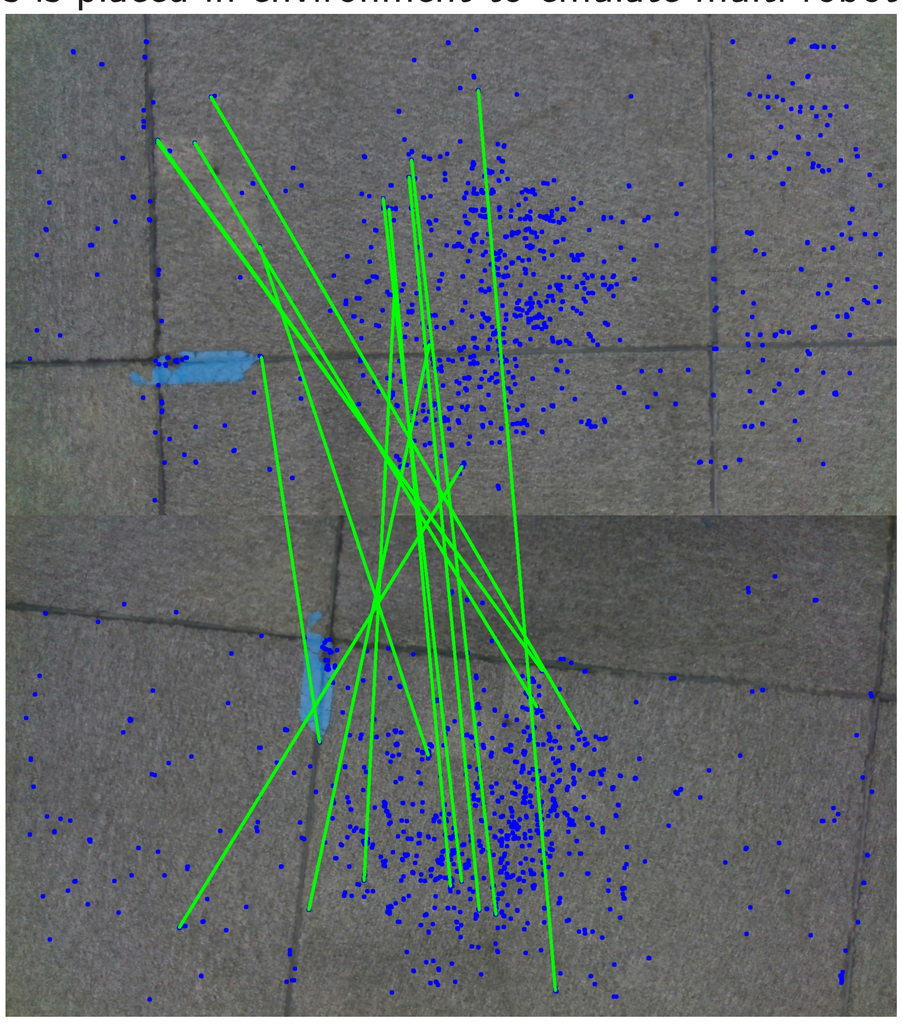
- No outward features
- No IMU or GPS
- Large operating area
- Many cooperative robots

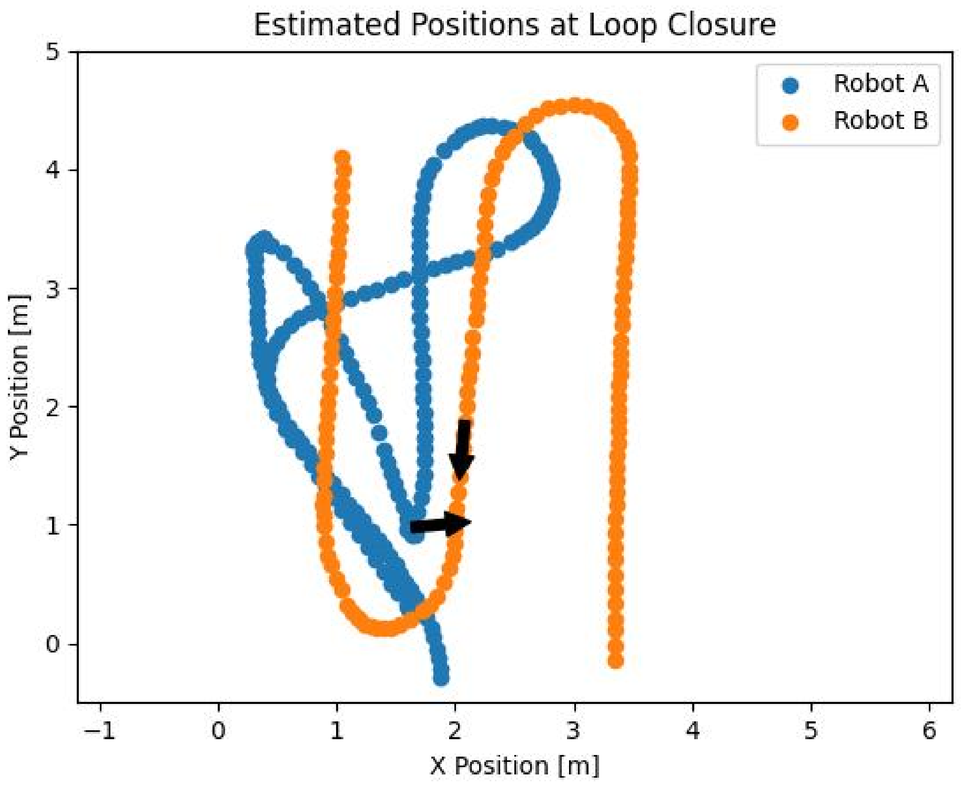
Architecture



Representative Data

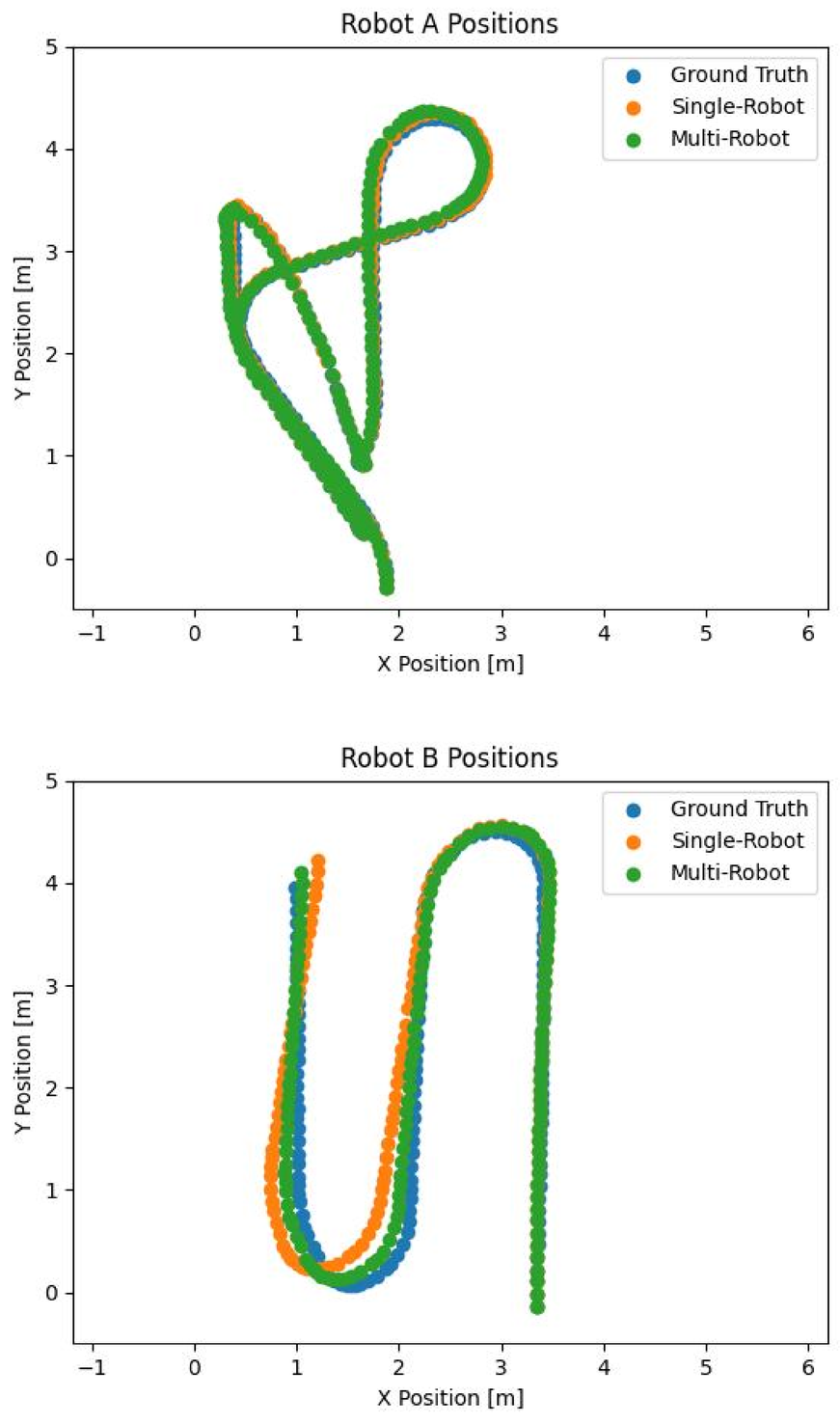
Robot with a downward-facing camera and motion capture markers is placed in environment to emulate multi-robot setup.





Results

Compare SLAM accuracy with vs without information sharing. Ground truth via motion capture allows high-precision evaluation. Information sharing permits Robot B to improve accuracy despite no self loop closures.



Robot	Method	RMSE Position $[m]$	RMSE Orientation $[rad]$
A	Single	0.031	0.007
A	Multi	0.043	0.017
B	Single	0.249	0.121
B	Multi	0.114	0.056

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